The Pyramid Puzzle

Whodunnit? · Triangles & Congruence · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Triangles & Congruence	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Triangle congruence rules (SSS, SAS, ASA, AAS, HL), corresponding parts and the reflexive property.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Ms. Okonkwo took it, in the Burial Chamber, using this method: Poured into a water jug.

The reveal (read aloud): Under cover of darkness, **Ms. Okonkwo** slipped into the **Burial Chamber**, hid the Golden Scarab inside a **water jug**, and carried it out as if collecting water. Every alibi and method you cleared pointed to her. Pharaoh-ly good work!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Dr. Amara · Place: Dig Site
2	Method: Hidden in a saddlebag
3	Suspect: Mr. Petrie · Place: Camp Tent
4	Method: Buried in the sand
5	Suspect: Zed · Place: Pyramid Entrance
6	Method: Wrapped in mummy cloth
7	Suspect: Prof. Lyle · Place: Gift Stall
8	Method: Disguised as a souvenir
9	Suspect: Hassan · Place: Oasis
10	Method: Packed in a camera case

Answer Key

The Pyramid Puzzle · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	In △ABC and △DEF: AB ≅ DE, BC ≅ EF and AC ≅ DF. Which congruence rule proves △ABC ≅ △DEF?	**B.** SSS	all three pairs of sides are congruent SSS	Dr. Amara was brushing sand from a vase at the Dig Site. Her whole team saw her.
2	In △ABC and △DEF: AB ≅ DE, ∠B ≅ ∠E and BC ≅ EF. Which rule proves the triangles congruent?	**A.** SAS	two sides and the angle **between** them SAS	The camel saddlebags were inspected. Only dates and bread inside.
3	In △ABC and △DEF: ∠A ≅ ∠D, AB ≅ DE and ∠B ≅ ∠E. Which rule proves the triangles congruent?	**C.** ASA	two angles and the side **between** them ASA	Mr. Petrie was filling out museum forms in the Camp Tent. The forms are signed and timed.
4	In △ABC and △DEF: ∠A ≅ ∠D, ∠B ≅ ∠E and BC ≅ EF. Which rule proves the triangles congruent?	**B.** AAS	two angles and a side that is **not** between them AAS	The sand around the camp was raked and sifted. Nothing was buried.
5	Two right triangles have congruent hypotenuses and one pair of congruent legs. Which rule proves them congruent?	**A.** HL	right triangles, hypotenuse and leg HL	Zed was guiding a tour group at the Pyramid Entrance. Twenty tourists remember him.
6	In △ABC and △DEF: AB ≅ DE, BC ≅ EF and ∠A ≅ ∠D. What can you conclude?	**D.** Not enough info	the angle is **not** between the two sides (SSA) Not enough info	The mummy cloths are all old and dusty. None was recently unwrapped.
7	△ABC ≅ △DEF. m∠B = (4x + 8)° and m∠E = 56°. Find x.	**C.** x = 12	corresponding angles are congruent (CPCTC) 4x + 8 = 56 x = 12	Prof. Lyle was buying a scarab keychain at the Gift Stall. The receipt has the time.
8	△ABC ≅ △DEF. AB = 3x − 2 and DE = 2x + 7. Find x.	**B.** x = 9	3x − 2 = 2x + 7 x = 9	The souvenir stalls are filled with cheap plastic beetles. The real Scarab isn’t among them.
9	△PQR ≅ △XYZ. Which angle corresponds to ∠Q?	**D.** ∠Y	the order of the letters matches: P↔X, Q↔Y, R↔Z ∠Y	Hassan was giving the camels water at the Oasis. A farmer saw him there.
10	Two triangles share the side AC. Which property justifies AC ≅ AC?	**A.** Reflexive	a segment is congruent to itself Reflexive	The camera case was opened at the gate. Just lenses and batteries.

Wrong-Answer Map

The Pyramid Puzzle · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	A. HL	Triangles are not stated to be right triangles	The water jugs were all poured out at noon. Only plain water ran out.
1	C. ASA	No angles are given	Dr. Amara was measuring the walls of the Burial Chamber. Her tape measure is still on the floor.
1	D. SAS	Needs an included angle, none is given	Ms. Okonkwo was photographing the carvings at the Pyramid Entrance. Her flash lit up the doorway.
2	B. SSS	Only two sides are given	Ms. Okonkwo was snapping pictures of the diggers at the Dig Site. The shots are time-stamped.
2	C. AAS	Only one angle is given	The big clay jug is cracked and leaks. Nothing could be stored in it.
2	D. ASA	Only one angle and two sides are given	Zed was telling spooky stories in the Burial Chamber. His whole tour group was there.
3	A. SAS	The given angles are not between two sides	Ms. Okonkwo was changing film in the Camp Tent. The flap was tied shut with her inside.
3	B. SSS	No third side is given	Every water jug was filled from the oasis well. All of them hold only water.
3	D. AAS	The side is between the angles, so it is ASA	Prof. Lyle was copying hieroglyphs in the Burial Chamber. His notebook has the proof.
4	A. HL	Not right triangles	The water jug has a very narrow mouth. Nothing bigger than a pebble could be poured in.
4	C. ASA	The side BC is not between the two angles	Ms. Okonkwo was bargaining at the Gift Stall for a postcard. The shopkeeper remembers her.
4	D. SAS	Only one side is given	Hassan was carrying water into the Burial Chamber for the workers. Wet footprints trail behind him.
5	B. SSS	Only two sides are given	Mr. Petrie was inspecting the Burial Chamber with a clipboard. The guard signed him in.
5	C. SAS	The right angle is not between the hypotenuse and leg	Ms. Okonkwo was taking sunset photos at the Oasis. The palm trees are in every shot.
5	D. Not enough info	HL applies to right triangles	The jugs were weighed at the gate. Every one weighed exactly what water should.
6	A. SAS	The congruent angle is not included between the sides	Ms. Okonkwo was adjusting her lens at the Pyramid Entrance. A tourist asked her for directions.
6	B. ASA	Only one angle is given	A jug was tipped over by the camp. The spilled water was clear, with nothing in it.
6	C. SSS	Only two sides are given	Zed was pointing out wall paintings in the Burial Chamber. The tour group clapped.
7	A. $x = 29$	Used supplementary: $4x+8=124$	The water jug is sealed with wax, and the wax is old and unbroken.
7	B. $x = 14$	Divided 56 by 4, ignored the 8	Dr. Amara was dusting off a statue inside the Burial Chamber. Her assistant held the lamp.
7	D. $x = 16$	Added 8 instead of subtracting	Ms. Okonkwo was photographing a sandstorm rolling across the Dig Site. The diggers took cover beside her.
8	A. $x = 7$	Ignored the -2	The camp’s jugs were measured and found far too tiny to hold a jeweled beetle.

8	C. $x = -5$	Sign errors on both sides	Prof. Lyle was translating a wall inscription in the Burial Chamber. The guard watched him.
8	D. $x = 5$	Sign error: $3x+2=2x+7$	Ms. Okonkwo was charging camera batteries in the Camp Tent. The generator hummed all night.
9	A. $\angle Z$	Matched the last vertex	Hassan was delivering lamp oil to the Burial Chamber. The workers signed for it.
9	B. $\angle X$	Matched by letter order of the alphabet, not position	Ms. Okonkwo was buying film at the Gift Stall. The receipt shows the time.
9	C. $\angle R$	Matched Q with the wrong vertex of the first triangle	The camp's water jugs were rinsed and left upside down to dry. None was used last night.
10	B. Symmetric	Symmetric reverses a statement $a=b$	Ms. Okonkwo was watching camels drink at the Oasis. She took a photo of the whole herd.
10	C. Substitution	Substitution replaces a quantity with an equal one	The jug on the supply cart has a hole in the bottom. It could not hold anything.
10	D. Transitive	Transitive chains two equalities	Mr. Petrie was counting display cases in the Burial Chamber. The tally sheet is dated.